Reproducing Yokum, Ravishankar and Coppock (2019)

A randomized control trial evaluating the effects of police body-worn cameras

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Archived at Zenodo: doi:10.5281/zenodo.24168425. This DOI always resolves to the latest release of the maintained version.

This repository reproduces the published results of Yokum, Ravishankar and Coppock (2019), a randomized trial of police body-worn cameras conducted with the Metropolitan Police Department of the District of Columbia. It contains a maintained rewrite of the analysis in

current R, a table that compares every checkable published number against both the deposited code and the rewrite, and a report on what does and does not reproduce.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

0.1 What is here

Table 1: Where the paper and its materials live. Both DOIs were resolved and checked against the registration agency's record.

Item	Reference
Article	https://doi.org/10.1073/pnas.1814773116
Supplementary appendix	https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.1814773116/-/DCSupplem
Deposited archive	https://doi.org/10.17605/OSF.IO/P6VUH
Preanalysis plan	Included in the deposited archive and reprinted as Appendix G
Errata	yokum_ravishankar_coppock_2019_errata.pdf, in this repository

```
yokum_ravishankar_coppock_2019/
  README.qmd                    this report, rendered to README.md and report/
  yokum_ravishankar_coppock_2019.Rproj
  run_all.R                     entry point: runs everything in order
  download_original.R           fetches and verifies the deposited archive
  original_manifest.csv         the deposit's file names, sizes and checksums
  original/                     the deposited archive (not redistributed here)
  maintained/                   the rewrite, one script per published float
  maintained/in_text_claims.R every quantity the article states in a sentence
  maintained/output/            everything the rewrite produces
  ground_truth/                 published values against archive and rewrite output
  ground_truth/published_claims.csv   the numeric-token extraction from the paper
  errata.qmd                    the corrections to the published text
  report/                       the PDF build of this report
```

To reproduce: clone the repository, open `yokum_ravishankar_coppock_2019.Rproj`, and run

```
source("run_all.R")
```

The first run downloads the deposited archive from OSF, which is 176 MB, and verifies every file against the checksums OSF publishes. Later runs skip files already present and correct. Each script under `maintained/` can also be run on its own.

1 Summary

Does the deposited archive run? Yes. All four deposited scripts execute at exit status 0 under R 4.6.2 and current package versions. The only complaints are a deprecation warning and a layout warning, both from the two coefficient plot scripts.

Does the deposited archive reproduce the paper? Almost none of it. The archive's one table-producing script fits every model on all 2,224 officers in the trial. The published tables report 1,922, the officers in the seven patrol districts, which is the sample the article says in its Methods section it analyses. Of the 280 published table values the archive's own output can be compared against, 227 disagree and 53 agree, and every one of the 53 agreements is a cell whose published value is zero or a standard error that happens to round the same way. `regression_tables.R` produces those numbers with no error, no warning and nothing on screen to suggest a problem.

Does the maintained rewrite reproduce the paper? Yes, on everything the deposit can support. Of 381 checkable published values, 379 agree to the precision the paper prints and 2 do not. A further 60 rows record published quantities that no deposit could reproduce, almost all of them because the officer covariates behind the covariate-adjusted results cannot be published without identifying individual officers. Beside the numeric comparisons, 17 claims the paper states in words are given a computed truth value; 14 hold and 3 do not.

Does the paper agree with itself? In 5 places it does not, and those are collected in `yokum_ravishankar_coppock_2019_errata.pdf` at the root of this repository as five entries. Two figure captions name a window an order of magnitude shorter than the one their figures cover, and because they carry the same error they are one entry; the rest are a compliance figure that disagrees with the article's own Table C.37, a pilot count that disagrees with its own Table A.1, and a cross-reference that points at a section the supplement does not have. None of them changes a conclusion, and none touches an estimate.

2 The paper

Yokum, David, Anita Ravishankar and Alexander Coppock. 2019. "A randomized control trial evaluating the effects of police body-worn cameras." *Proceedings of the National Academy of Sciences* 116 (21): 10329 to 10332.

Roughly half of 2,224 eligible Metropolitan Police Department officers were randomly assigned a body-worn camera, blocked on district and on officer characteristics within district. Outcomes came from administrative records over matched 212-day windows before and after each district's camera rollout, and were expressed as yearly rates per 1,000 officers. The paper reports two estimators: a difference in means weighted by the inverse probability of assignment, and a regression that adds the pretreatment value of the outcome and officer covariates. The main

analysis is the seven patrol districts, 1,922 officers, excluding the station detail and the special units.

One small discrepancy inside the paper is worth noting before anything is compared to it. Eq. 1 in the Methods section writes the covariate-adjusted specification with a vector of major-block indicators alongside the pretreatment outcome and the officer covariates. None of the sixteen published covariate-adjusted tables lists a block indicator: each prints the pretreatment outcome, gender, a three-category race variable and length of service, and nothing else. The inverse probability weights already account for the differing assignment probabilities across blocks, so the omission is defensible, but the equation and the tables do not describe the same regression.

The headline result is a null. Across all 45 measured outcomes, covering use of force, civilian complaints, policing activity and judicial outcomes, cameras changed nothing detectably.

3 The deposited archive

The deposit at osf.io/p6vuh holds 11 files in two folders, 176 MB in total.

Table 2: The deposit, as listed by the OSF API.

File	Bytes	OSF id
Pre-Analysis Plan (PAP)/Pre-analysis Plan - MPD BWC RCT - v1.pdf	408,266	hpmrt
Replication Data/BWC_dv_df.rdata	1,855	kmjxg
Replication Data/README.txt	593	4pzyt
Replication Data/coefplot_all_dvs.R	1,914	vswpu
Replication Data/coefplot_main_dvs.R	1,920	f2c93
Replication Data/day_level_anon.csv	177,902,506	h9wpc
Replication Data/dv_df.csv	8,985	ctyb9
Replication Data/officer_level_anon.csv	1,865,518	gsbwt
Replication Data/regression_tables.R	15,972	muqpj
Replication Data/time_series_plot.R	3,292	spr5g
TheLabDC_MPD_BWC_Working_Paper_10.20.17.pdf	4,650,695	rk2pa

`download_original.R` fetches each file by its OSF identifier and checks it against both published hashes. All 11 files verify on MD5 and on SHA-256, and the local copy is byte for byte what OSF serves. The script also refuses to continue if `original/` holds anything the manifest does not list, so a stray file left behind by a previous run cannot quietly become part of the archive.

Two notes on the shape of the deposit. It carries a preprint of the paper from 2017, not the published article, so the published values used below were read from the article and its appendix rather than from anything in the deposit. And the deposit's own `README.txt` lists five R scripts where four are present: `regression_tables_cov.R`, the script that would have produced the covariate-adjusted tables, is named but absent.

3.1 What the deposited scripts do

Table 3: The deposited scripts and the published floats they correspond to.

Script	Produces	Status
coefplot_main_dvs.R	Figure 1	Runs, two warnings
coefplot_all_dvs.R	Figure C.1	Runs, nine warnings
regression_tables.R	Tables C.5 to C.37, odd numbered	Runs clean
time_series_plot.R	Figures E.3 and E.4	Runs clean
regression_tables_cov.R	Tables C.6 to C.36, even numbered	Absent from the deposit

Everything that is present runs, all four at exit status 0. `coefplot_main_dvs.R` and `coefplot_all_dvs.R` raise a deprecation warning for `geom_errorbarh()` and then one layout warning from `position_dodgev()` per panel drawn, which is one warning for the single-panel figure and eight for the faceted one. Nothing errors. No script saves an output either: all four draw to the screen or print to the console, so a reader who runs them has nothing on disk to compare against anything. Run non-interactively each one leaves an `Rplots.pdf` in the working directory, which is the only reason to care where they are run from, and it is why the archive here is copied out before it is run rather than executed inside `original/`.

3.2 Three ways the deposited code departs from the paper

The sample. Every model in `regression_tables.R` is fit on the full 2,224 officers. Every published table reports 1,922. The two coefficient plot scripts have the same problem. This single difference is why the deposited output does not match the paper anywhere, and it is invisible at runtime.

Table 4: Table C.5, the first of the appendix tables. Standard errors in parentheses. The deposited script's numbers come from all 2,224 officers; the published table and the rewrite use the 1,922 in the seven patrol districts.

Outcome	Published	Deposited script	Rewrite	N, deposited	N, rewrite
Use of Force	73.6 (87.0)	69.8 (76.6)	73.6 (87.0)	2224	1922
Use of Force (Serious)	13.8 (14.1)	8.4 (12.7)	13.8 (14.1)	2224	1922

Use of Force (Other)	59.8 (83.4)	61.4 (73.4)	59.8 (83.4)	2224	1922

The second estimator in the figures. Both coefficient plot scripts build a vector of covariate-adjusted formulas and then never use it: the object named for the adjusted fits is computed from the unadjusted formulas instead. The figures the deposited scripts draw therefore plot the same estimates twice, one series labelled as though it were adjusted. The rewrite uses the adjusted formulas, which is plainly what the deposited code intends.

The adjustment itself. Even with that corrected, the adjusted series cannot be the published one. Every published covariate-adjusted table conditions on officer gender, race and length of service, and the deposit contains none of the three, and says so, because releasing them would identify individual officers. The deposited scripts substitute an adjustment on the pretreatment outcome alone and note the substitution in a comment. The rewrite keeps that reduced adjustment and labels it for what it is. No published number from any covariate-adjusted table or figure series can be checked against this deposit, by anyone.

4 The maintained rewrite

`maintained/` holds one script per published float, a shared `helpers.R` that defines the two estimators and the analysis sample, and two cleaning scripts. Output goes to `maintained/output/` and is committed, so a fresh run can be diffed against it without downloading anything.

Table 5: The rewrite.

Script	Produces
helpers.R	packages, paths, the analysis sample, the two estimators
clean_officer_level.R	the officer file, whole and cut to the seven districts
clean_day_level.R	the officer-day panel collapsed to district by period rates
table_a1_random_assignment.R	Table A.1
table_a4_deployment_dates.R	Table A.4
table_c5 ... table_c37 (17 scripts)	the odd-numbered appendix tables, C.5 through C.37
figure_1_main_outcomes.R	Figure 1
figure_c1_all_outcomes.R	Figure C.1
figure_e3_use_of_force_time_series.R	Figure E.3
figure_e4_complaints_time_series.R	Figure E.4
text_main_estimates.R	the three effect estimates the Results section states in words
text_sample_and_compliance.R	the sample counts and the compliance means in Methods
text_null_results_check.R	a count of outcomes reaching significance
text_district_by_district.R	the seven district-level estimates the Discussion asserts

Substitutions from the deposited code: `%>%` becomes the native pipe; `geom_errorbarh()` becomes `geom_linerange()`; `position_dodgev()` from the unmaintained `coefplot` package becomes `position_dodge()` with the outcome on the discrete axis; `lm()` with `starprep()` becomes `lm_robust()`, whose default standard error is already HC2; `stargazer()` printing LaTeX to the console becomes a CSV at full precision plus a rounded LaTeX table written to a file; and `rm(list = ls())` at the top of every script is gone, since the scripts assume nothing about the session they start in.

Two smaller decisions are worth recording. The weights column is pulled out of the data frame before it reaches `lm_robust()`, because a weights argument is evaluated inside the data first and the column is called `weights`. And district labels are read from CSV as text on purpose: `readr` accepts the Fortran double exponent, so a guessed column of `1D` through `7D` comes back as the numbers 1 through 7.

4.1 Coverage

Every published float has a row in the ground truth and a count of what it prints in the extraction. Five are recorded as having no source in the deposit at all:

- **Tables A.2 and A.3**, the pretreatment distributions of officer race, sex and length of service. Those covariates are not deposited.
- **Tables C.6 through C.36, even numbered**, the sixteen covariate-adjusted tables. Same reason, and the script that made them is missing besides.
- **Figure D.2**, all outcomes under the alternate measurement strategy, which uses each district's full observation window. The deposited officer file carries only the 212-day rates.
- **Figure E.5**, calls for service against videos uploaded. The calls-for-service series is not deposited.
- The **adherence figures** in the Discussion, 98 per cent of days and 96 per cent average adherence, which rest on the same calls-for-service data.

5 Ground truth

`ground_truth/build_ground_truth.R` assembles 441 rows. Each carries the published value, typed from the article or the appendix and used only as a comparison target; the value the deposited specification produces, read from a file written by `ground_truth/archive_estimates.R`; and the value the rewrite produces, read from `maintained/output/`. No published number is an input to any computation anywhere in this repository.

Table 6: Ground truth, all rows. A row is not checkable when the paper states no number, or when nothing in the deposit can produce one.

Comparison	Agree	Disagree	Not checkable
Deposited specification against the paper	53	227	161
Maintained rewrite against the paper	379	2	60

Table 7: Checkable rows by where the number appears.

Where the number appears	Rows	Rewrite agrees	Rewrite disagrees
Appendix Tables A.1 and A.4	43	43	0
Figure captions and plotted counts	9	9	0
Appendix Tables C.5 to C.37	280	280	0
Quantities stated in sentences and table notes	49	47	2

Every one of the 280 cells the odd-numbered appendix tables print reproduces exactly, as do all 36 cells of Table A.1 and the seven of Table A.4's ten deployment dates that the deposited officer-day panel covers.

Table 8: The rows where the paper disagrees with itself. Each is corrected in the errata.

Where	Claim	Paper says	Data
Figure E.3 caption (p. 37)	Days before and after deployment the figure covers	90	FALS
Figure E.4 caption (p. 38)	Days before and after deployment the figure covers	90	FALS
Methods (p. 2)	Videos per year, officers assigned a camera	665	663.1
Appendix A (p. 3)	Pilot officers not given cameras	180	178.0
Appendix A.1 (p. 5)	The alternate measurement plots are said to be in Section 4	-	FALS

The compliance sentence in Methods is the sharpest of them. The article says treatment officers uploaded "about 665" videos a year. The weighted mean in the analysis sample is 663.1, which is also what the article's own Table C.37 implies, since its constant of 13.9 plus its coefficient of 649.1 is 663.0. No sample or weighting scheme available in the deposit returns 665. The gap is two videos out of 663 and changes nothing, but the sentence and the table it summarizes do not agree with each other.

6 The extraction and the two instruments

The ground truth answers "does this published number reproduce?" It cannot answer "is this published number in the ground truth at all?", and that second question is where the errors in

this article turned out to live. Two further files answer it.

`ground_truth/published_claims.csv` is the extraction: every numeric token in the article and in its supplement, read line by line rather than searched for, classified by hand into the five kinds a claim can be. It carries 191 rows. `pipeline` claims are quantities the analysis produces, `descriptive` claims are assertions about shape, sign or count that need a truth value rather than a number, `definitional` claims are scale endpoints and design constants, `structural` claims are page furniture and float inventories, and `transcribed` claims are numbers copied from another document and unable to move with the pipeline.

Table 9: The extraction from the article and its supplement.

Kind of claim	Rows	Need a block
definitional	45	35
descriptive	23	23
pipeline	27	27
structural	84	4
transcribed	12	0

The coverage boundary is stated rather than assumed. Inside it are the article's body, from the abstract through the Discussion, and supplement sections A through E, including every table body, figure caption and table note. Outside it are three classes, each recorded in the extraction as a row of its own: the numbered reference list and the bracketed citation markers in the body; supplement Appendix F, which reproduces MPD General Order SPT-302.13 verbatim; and supplement Appendix G, which reprints the preanalysis plan as a separately paginated document. The last two are transcriptions of prior documents, so none of their numbers can move with the pipeline. Page numbers, ZIP codes, statute numbers, disposition codes, an IRB protocol number and the DOI are recorded as structural rows and are not claims about anything the data could contradict.

The extraction also counts what each published float prints, which is what makes partial coverage visible.

Table 10: What each published float prints, and how much of it the ground truth compares. The two coefficient plots print no estimate on their faces, so their count is what they plot.

Published float	Floats	Numbers published	Covered	Reproduced by the r
Appendix Tables A.1 and A.4	2	46	46	
Appendix Tables A.2 and A.3	2	20	0	
Figures 1 and C.1	2	96	48	
Appendix Tables C.5 to C.37, odd numbered	17	280	280	

Appendix Tables C.6 to C.36, even numbered	16	706	0
Figures D.2, E.3, E.4 and E.5	4	90	0

374 of the 1238 numbers the published floats carry are compared cell by cell. The rest are the sixteen covariate-adjusted tables, the two pretreatment covariate tables and the alternate-strategy figure, all of which rest on officer covariates the deposit cannot contain.

`maintained/in_text_claims.R` is the second instrument. Every claim the extraction marks as needing a block, 89 of them, is recomputed there from `maintained/output/` beside the sentence that states it, and printed as a labelled line. It reads the extraction, so that a block can name the article's own precision, and it never reads the ground truth, because agreeing with the comparison would prove nothing. Where `build_ground_truth.R` reaches a quantity through one of the `text_*.csv` summaries, this file goes back to the table or the cleaned data that summary was built from, and where the build goes through a table, the claims file goes through the summary. The two derivations are separate on purpose: where they disagree, one of them is wrong.

`build_ground_truth.R` runs the coverage gate, in this order. Checks that depend only on the extraction come first, so that a wrong precision trips its own check rather than the value comparison downstream: identifiers are unique, claim types and comparison operators are drawn from the allowed sets, every `pipeline` and `descriptive` row is marked as needing a block, and every stored published value survives a round trip through the number of decimals recorded beside it. The locus rule follows in all three of its states: a row where either verdict is adverse must name where the fault lies, a clean match must not, and a row with no verdict may. The extraction is then reconciled against the ground truth's own transcription of the same pages, and the float inventory against what the ground truth covers. Last, the claims file is run non-interactively into its own environment, its output is captured, and the printed claim lines are counted: 89 printed against 89 required, with the identifiers equal in both directions, and each printed value compared against the ground truth's at the precision the page itself uses.

The gate was tested by breaking it three ways. Deleting a block fails the count. Corrupting a value in the claims file fails the cross-instrument comparison. Corrupting a precision in the extraction fails the round-trip check, and fails it before anything else reads that precision.

7 Errata

The five corrections to the published text are collected in `yokum_ravishankar_coppock_2019_errata.pdf`, built from `errata.qmd` at the root of this repository with every corrected value computed at render time from `maintained/output/`. That file also writes `errata_entries.csv`, the spine every entry is defined in and the source of the count above. None changes a conclusion and none touches an estimate.

1. The captions of supplementary Figures E.3 and E.4 say the figures cover 90 days before and after deployment. They cover 690 days before and 510 days after. The deposited `time_series_plot.R` applies a 90-day filter to a pooled version of each figure that the supplement does not print; the district-by-district version it does print carries no such filter.
2. The Methods section says treatment officers uploaded about 665 videos a year. The figure implied by the article's own Table C.37, and produced by the analysis sample, is 663.
3. Appendix A says 180 officers in the pilot were not given cameras. The article's own Table A.1 gives 79 in 5D and 99 in 7D, which is 178. The treated count in the same sentence, 325, is exactly right.
4. Appendix A.1 says the alternate measurement plots are "provided in Section 4." The supplement's sections are lettered A through G; the plots are Figure D.2, in Appendix D.

This is separate from the corrections the rewrite makes to the deposited code, which are described above and are errors in the archive rather than in the article.

8 Figures

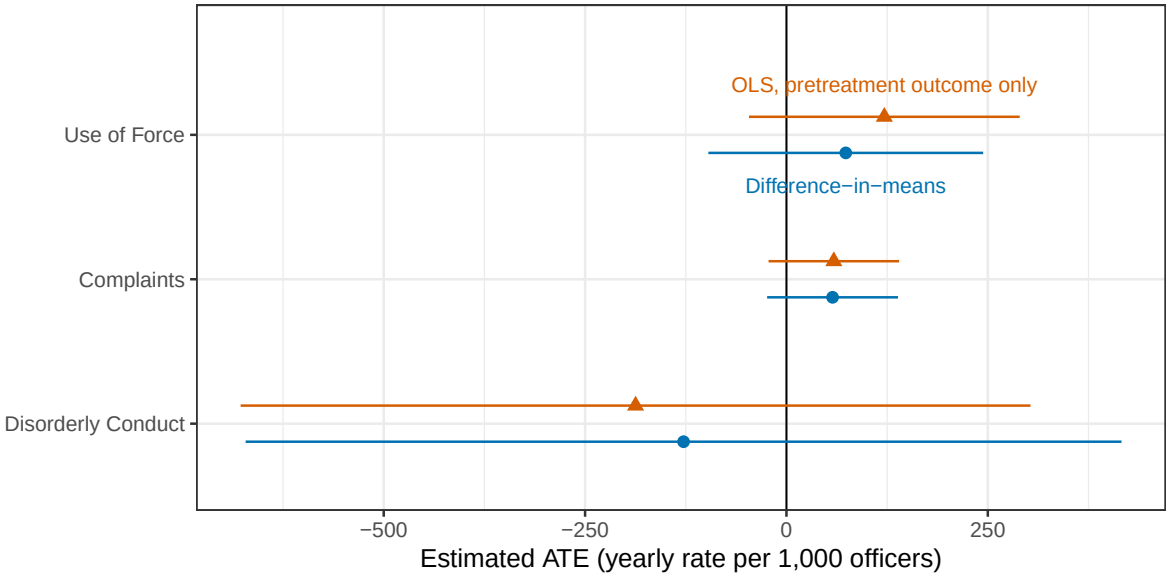


Figure 1: Figure 1 of the article. The circles are the published difference-in-means estimates. The triangles are not the published second series: the deposit cannot support the covariate adjustment the paper used, so this adjusts for the pretreatment outcome alone.

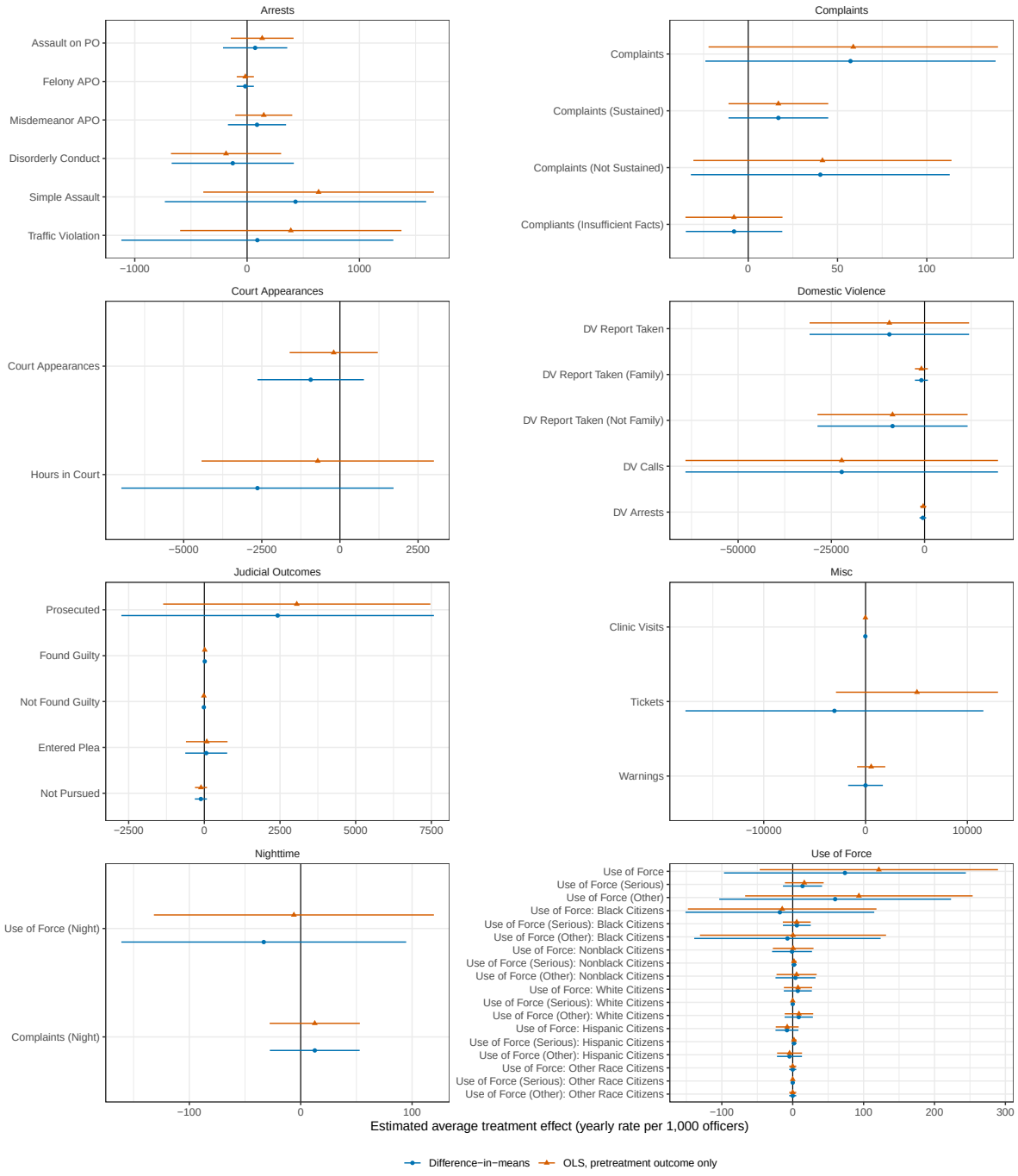


Figure 2: Appendix Figure C.1, all 45 outcomes. The same caveat applies to the adjusted series.

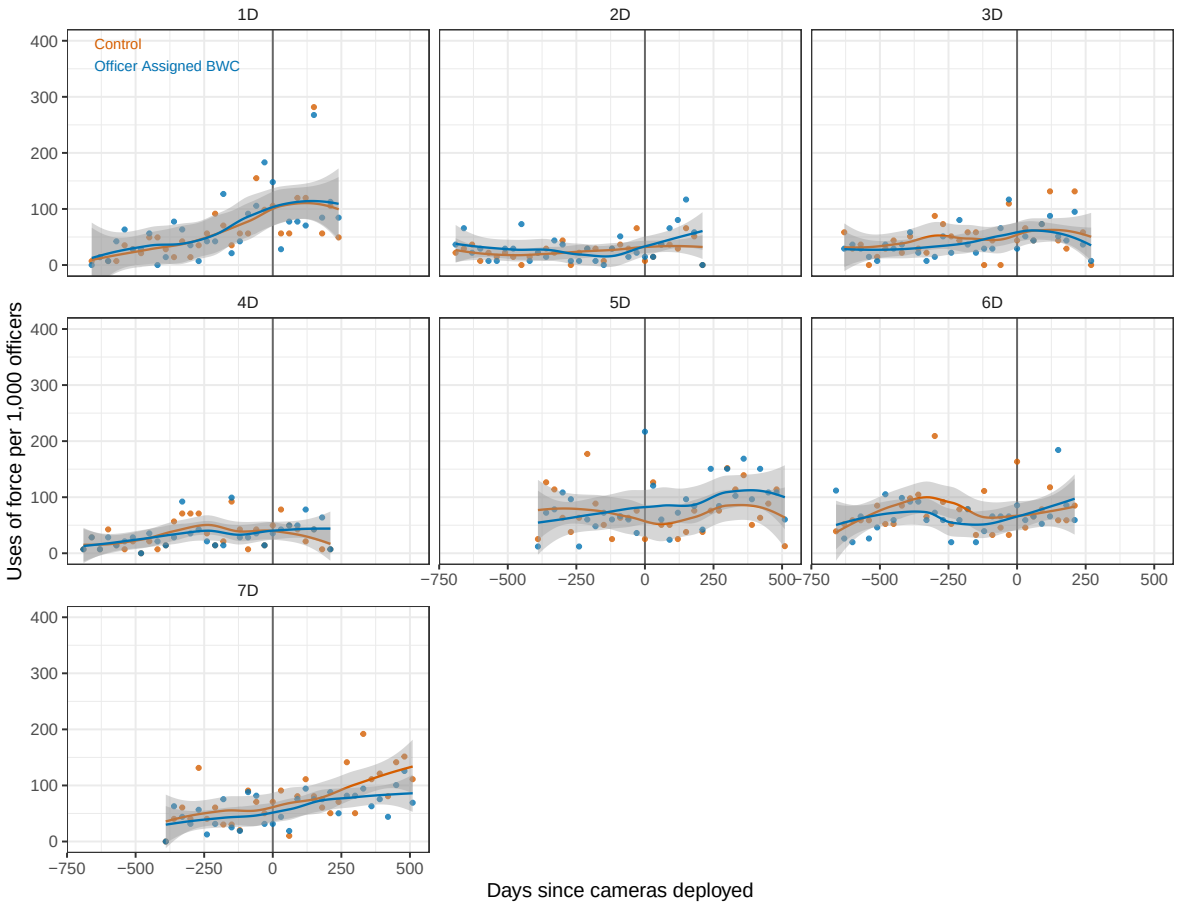


Figure 3: Appendix Figure E.3, uses of force per 1,000 officers by district and 30-day period.

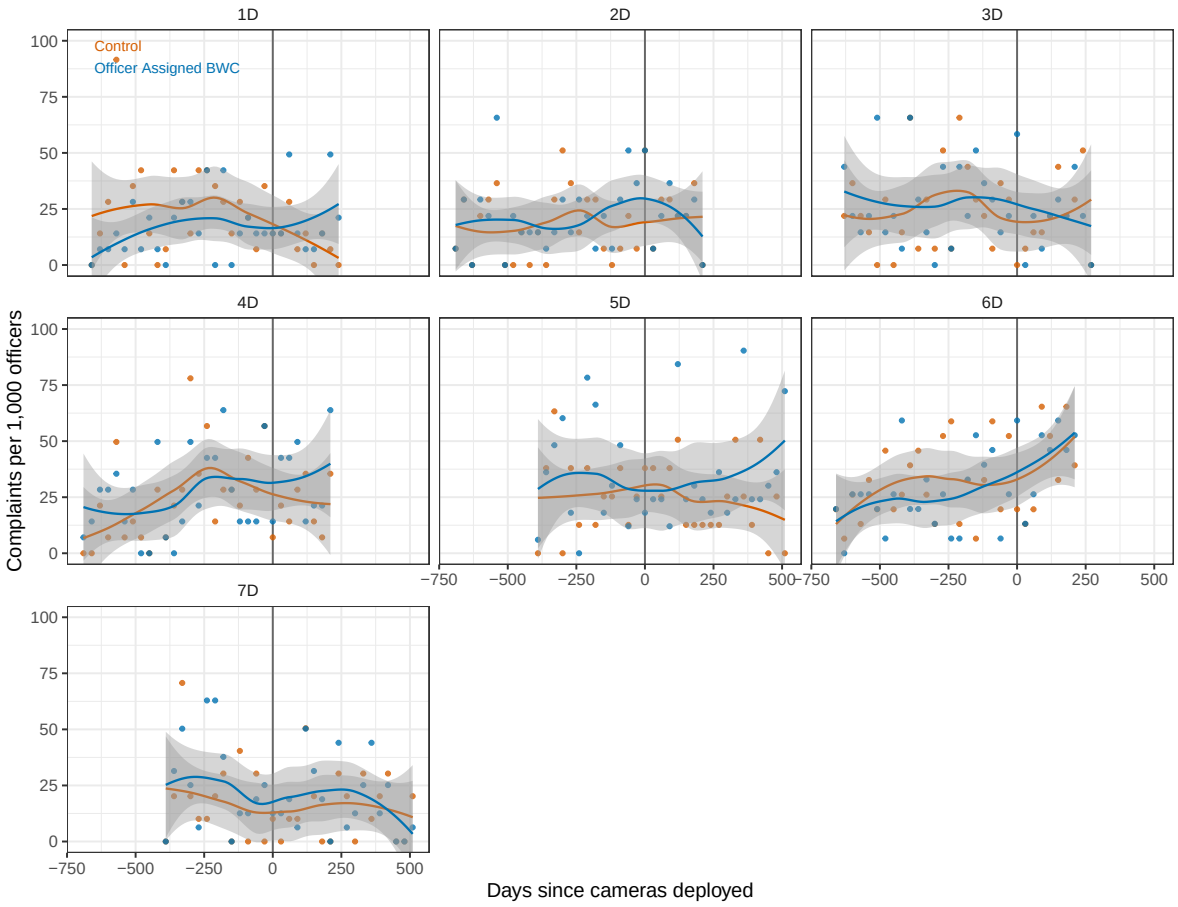


Figure 4: Appendix Figure E.4, complaints per 1,000 officers by district and 30-day period.

None of the published figures prints a number, so each figure script writes a CSV of the values it plots and the ground truth compares those. Figure C.1 plots the same unadjusted estimates the appendix tables print, and across 45 outcomes, matched on the outcome column name rather than the printed label, the largest absolute difference between the figure's values and the tables' is 0. Figures E.3 and E.4 plot 432 district by period rates each, reproduced from the deposited officer-day panel.

One caption does not describe its figure. Figures E.3 and E.4 are captioned "90 days before and after BWC deployment" and say there is no significant difference "in either the 90-day period before or after". The published axis runs from 750 days before deployment to 500 days after, and the underlying panel covers 690 days before and 510 after. The window in the caption is not the window in the plot, and the correction is the first entry in the errata.

9 Verification

Every script runs in a clean session and the whole pipeline runs end to end from `run_all.R`. Running it twice returns byte-identical output, every file included: R's `pdf()` device stamps the wall clock into each figure's `/CreationDate` and `/ModDate`, and `run_all.R` blanks those two fields after the last analysis script, so the determinism check covers the figures rather than exempting them. No estimator used anywhere here draws a random number, so there is no seed to set and no dependence on which sampler the running version of R provides.

`run_all.R` begins by sourcing `download_original.R`, so every run re-verifies `original/` against the manifest before any analysis reads from it, and stops if a file has changed or if a file the manifest does not list has appeared. The deposit's own scripts are never executed inside `original/`, which is what a checksum gate placed after the fact would fail to catch.

Table 11: R environment. Rendered under R version 4.6.2 (2026-10-29).

Package	Version
tidyverse	2.0.0
estimatr	2.0.1
broom	1.0.13
ggplot2	4.0.3
readr	2.2.0
dplyr	1.2.1
purrr	1.2.2
knitr	1.52
kableExtra	1.4.1
here	1.0.2
openssl	2.5.0